Enhancing Human-Computer Interaction: An Accurate Video-Based American Sign Language Translation Model Using Machine Learning Techniques

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Abstract: American Sign Language (ASL) is the primary communication method used by individuals who cannot speak or hear well. Only about 2.8% of the United States population is proficient in ASL, making communication challenging and inconvenient for people who use ASL. To address this issue, we have developed a model that processes video data and translates ASL fingerspelling signs into English letters. We evaluated several approaches to achieve this goal, with the most effective method involving splitting the video into frames, followed by using Google's MediaPipe Library to extract information on the locations of hand landmarks in each image. The extracted points were then normalized and fed into a classification model. The models tested include Random Forest, XGBoost, and Logistic Regression. The training phase yielded accuracies of 98% for the Random Forest model, 97% for the XGBoost model, and 94% for the Logistic Regression model. However, the Logistic Regression model outperformed the other models for generalization. Our model demonstrates robust performance despite variations in hand color, camera position, background, and other potential distractions, noted as roadblocks in prior research. This work contributes to enhancing Human-Computer Interaction for individuals who use ASL.

Keywords: American Sign Language (ASL), Gesture recognition, Feature extraction, Support vector machines(SVM), Human-computer interaction (HCI), Sign language interpretation, Machine learning models, Random Forest, Logistic Regression, XG Boost

I. INTRODUCTION

With the increase of computers in the modern world, the interactions between computers and humans, specifically touchless interactions[1,2], have become very prominent. Due to the pandemic, there has been an upward trend of touchless human-to-computer interactions[3]. We also see many advancements that have been made because of deep learning techniques[4,5]. Interactions with sign language rather than using a mouse and keyboard also tend to seem more natural. There are two main ways of touchless interactions - sound recognition, and gesture recognition. This paper focuses on providing people who cannot use sound recognition, with a reliable model to recognize images of the American Sign Language.

ASL helps people with hearing and speech impairment communicate. There are about 10,000 signs that correspond to English words or phrases. These signs consist of hand gestures, hand movements, and facial expressions. To recognize these gestures, expressions, and movements, both the hands and the face have to be analyzed and recognized. ASL, like most other languages, has its own syntax[7], which makes direct translation very difficult. Instead, our model currently concentrates on identifying the different signs used for fingerspelling. Fingerspelling (Fig 1) is also used as an alternative to communicate words that do not have signs.

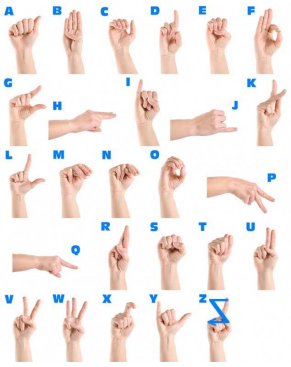


Figure 1 Fingerspelling and corresponding English Alphabets

Fingerspelling is the act of spelling words using the 26 signs that represent the 26 letters of the English alphabet. In the future, other signs of ASL can be added, such as using gestures as commands. Some other practical applications of this work could be smart appliance control, traffic light sign recognition, and recognizing other sign languages.[9,10]

In recent years there have been many approaches to make a machine "learn" the American Sign Language.

Starner et al.[11] used wearable devices to track hand movements, and Ye et al.[12] used a 3D recurrent convolutional neural network (3DRCNN) to predict gestures given a continuous video. Each approach has its benefits and downsides, which makes choosing a specific approach, difficult and time-consuming.

Ineffective Efforts/failed attempts

Initially, we tried a simple sequential convolutional neural network(CNN) with 12 layers. Before feeding the data to the model, we resized the images to size 32 pixels by 32 pixels. We were able to achieve a 96% accuracy on the validation dataset. However, We did not get good results when using the model to predict images of my own hand. We increased the image size to 128 pixels by 128 pixels so that the model has more data that it could learn from, but this too did not generalize well. Realizing that we were heavily restricted by our small dataset size, we decided to try transfer learning. This approach was similar to the one used by Avina et al.[13]. However instead of using Resnet50 as our base model, we used MobileNet. We trained our model with the help of MobileNet, but it still did not generalize. To try and increase the size of the dataset and introduce variation into the dataset, we tried different methods of augmentation - we tried to rotate images around the origin. This however confused the model and it did not train well. Then we tried to add noise to the images so that it would work better in different settings with different hands. This gave us better results than we had seen before. The model was able to recognize the sign we were holding up 60% of the time. We still did not have enough data to train a more complex model. We then shifted our approach to a two-step method to tackle the problem, namely segmentation and classification.

II. Our Approach

Segmentation involves getting the region of interest (ROI). This reduces the amount of data needed for the classification model and provides data with no distractions. This approach also tends to generalize better than just passing the raw image. Chang et.al.[14]. tested the use of segmentation by comparing the results of segmented data to other types of data. They observed a notable difference in accuracy between the classification results of passing segmented images vs raw images, the difference being a 13 percent increase in accuracy. This shows the benefit of segmentation. But, they realized that the differences in light, background, and angles were posing a big problem in getting the segmentation right. A similar approach was taken by Feng et al.[15]. Their goal was to try and recognize 9 different hand gestures by taking the 2-step approach: feature extraction and classification. They used the Histogram of Oriented Gradients(HOG) feature descriptor to segment out the background and make classification easier. Then they used an SVM algorithm to classify the different gestures. Due to the limitation of time and resource consumption, they were restricted to recognizing only 9 hand gestures. To overcome these limitations, we decided to use MediaPipe as our segmentation model.

Using Mediapipe, allowed us to save a lot of time and resources while benefiting from the segmentation. Using MediaPipe is simpler because, instead of passing 120,000 parameters to the classification model, we pass approximately 100 parameters with a dataset that has been feature-engineered.

We also came across another popular pose detection model called OpenPose. We chose Mediapipe over OpenPose due to two reasons. Firstly, OpenPose is more suited for general pose recognition due to limiting the amount of landmarks on the hand or palm area. Second, OpenPose is more computationally intensive and requires a lot more resources than MediaPipe. These reasons, as well as the extra features and support that come with MediaPipe, led us to use MediaPipe.

As opposed to the previous attempts, our approach to recognizing fingerspelling involves a 2 step process: segmentation using MediaPipe and classification. The first step in getting the important features was to run the images through MediaPipe, which is a library made by Google that takes an image of a hand and returns points(x, y, and z) of specific landmarks that represent joints on a hand. Then after a few methods of feature engineering, specified in the methodology section, we end up with around 100 features which are further used to train a classification model.

III. Methodology

Data Collection

The dataset used in this research was sourced from Kaggle, a platform where programmers share code and datasets[16]. We selected this dataset because it contains 3,000 images for each letter of the alphabet, with each image being 200x200 pixels. The images were captured under various conditions and lighting to introduce variability, aiding the model's learning process. However, the dataset only includes images of right hands, limiting the model to predicting right-hand signs.

Preprocessing

All images were processed using Google's Mediapipe hand recognition library, which identifies 21 landmarks per hand (Fig 2). Each landmark is represented by a normalized x and y values, with the z value scaled similarly to the x and y value. To simplify the sign identification process, the z values were disregarded.

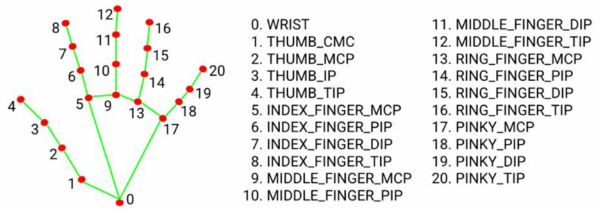


Figure 2 Mediapipe returns 21 tuples of landmarks per hand image

To make the data simpler for the model to understand we used several different strategies to extract more features and removed some unnecessary ones.

Distance Calculation: The first strategy involved, calculating the distance between each landmark and the bottom of the palm (landmark 0). The distance was calculated using the distance formula:

$$D = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Fig 3 shows the distance calculation between landmarks 0 and 17. This method aimed to help the model recognize differences in landmark positions. However, this alone did not give the model enough data to accurately predict the signs.

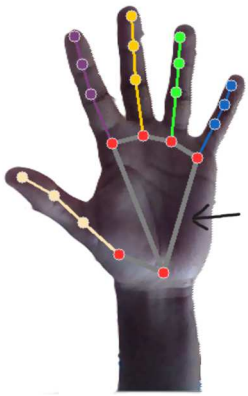


Figure 3 Distance between two landmarks

Angle Calculation: The second strategy added the angle between landmark 0 and every other landmark as a feature. Fig 4 shows the angle between landmarks 0 and 17. The angle was calculated using the formula:

$$\theta = arctan((y_2 - y_1)/((x_2 - x_1) + 0.000001))$$

After that, we would do a check to see if the width value was below 0. If it was then, we would add pi because of the range limitations of arctan.

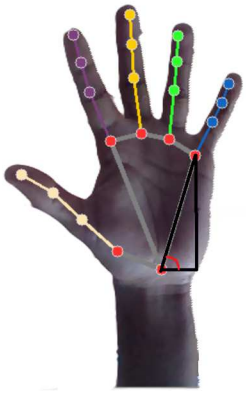


Figure 4 Angle between landmarks

The distance and angle calculations worked well but the model still could not differentiate between similar signs. Signs like N & M have minimal differences that make it hard for the model to differentiate them. Some examples of similar signs are shown in Fig 5.

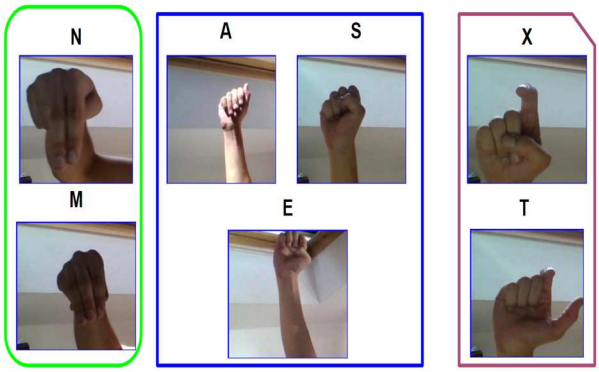


Figure 5 Similar Signs

Fingertip Distance Calculation: To address the difficulty of distinguishing between similar signs, we calculated the distances between the fingertips. We used the distance formula to calculate the distance between 2 landmarks. Fig 6 shows an example of the distances calculated for a certain hand.

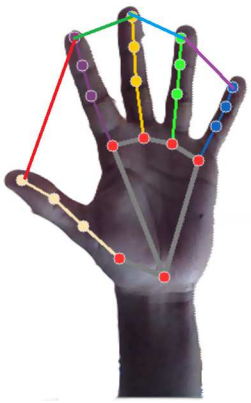


Figure 6 Fingertip differences

IV. IMPLEMENTATION

Training

After obtaining the data we moved on to training. First, we tried a Logistic Regression model with 1000 iterations. We achieved a 94% test accuracy. Then we tried xgboost with the parameters: objective="binary:logistic", colsample_bytree=0.5, gamma=0, learning_rate=0.1, max_depth=3, reg_lambda=0, scale_pos_weight=1, subsample=0.8. We got these parameters by using GridSearchCV, which is a library that runs the model on various combinations of parameters to find the best one. It achieved a 98% test accuracy. Lastly, we tried a Random Forest model with a depth of 100 which gave us a 99% test accuracy.

Fig 7 shows the overall process taken to arrive at the results presented in this paper.

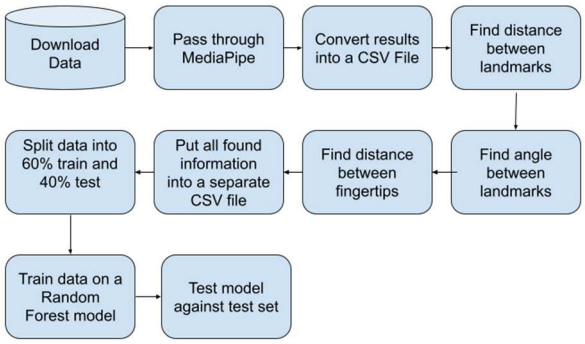


Figure 7: Flowchart of methodology

V OBSERVATIONS

Fig 8 shows each model's accuracy on the test data. The test data was obtained by splitting from the original dataset before training the model. All the models performed well, but the Random Forest model was the best out of them, showing a 99.01 percent accuracy on the test data.

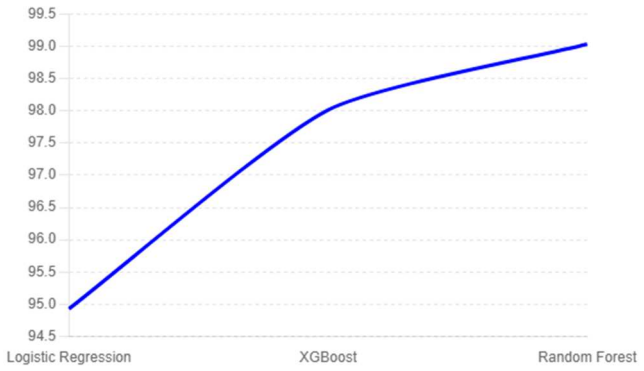


Figure 8: Accuracy of overall different models

However, to make sure that the model was not overfitting the given data, we made our own dataset. This dataset consisted of 1000 images per sign. We then tested each model on these images. The results are shown in Fig 9.

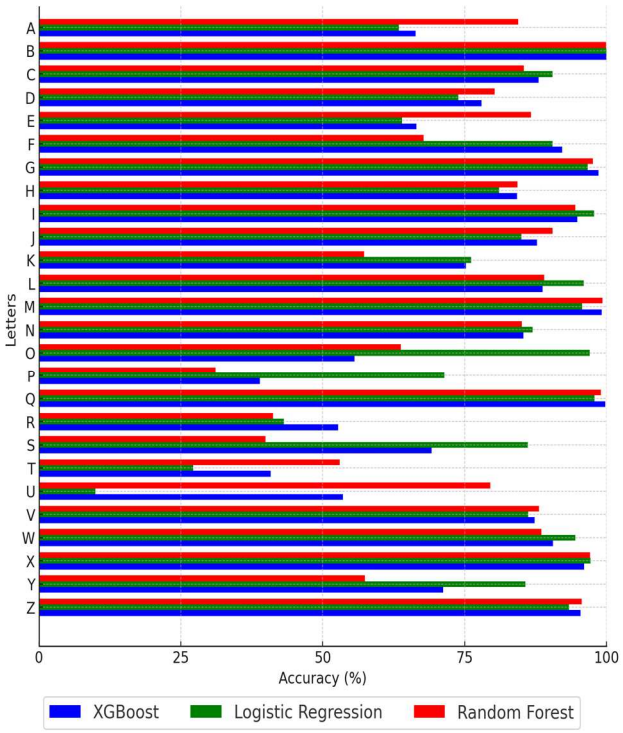


Figure 9: Accuracy of different models for each letter

Fig 9 shows the accuracy for each model on each alphabet. The green represents the Logistic Regression model, the blue represents the XGBoost model, and the red represents the Random Forest model. The average accuracy for the Logistic Regression, XGBoost, and Random Forest models were 80.23, 79.02, and 78.32 respectively. This shows that the Logistic Regression model was the best at generalizing even with the lower accuracy on the validation set.

VI. CONCLUSION

In this study, we successfully developed a model that translates American Sign Language (ASL) gestures into English text, addressing the communication challenges faced by the ASL community. The integration of MediaPipe allowed us to overcome limitations associated with dataset size and computational resources, streamlining the feature extraction process through the use of key hand landmarks. This approach not only reduced the complexity of our models but also improved their robustness in real-world applications by removing the limiting factors of differences in background, skin color, distractions, and many other factors. The feature engineering we did after segmentation differentiates our project from various others. It also allows the model to be trained efficiently and not require high-level model archetypes.

Among the models tested, the Random Forest model demonstrated the highest accuracy on our validation dataset, while the Logistic Regression model showed superior generalization capabilities on new data. Specifically, we achieved an accuracy of 99 percent for the Random Forest model while testing with the validation dataset, and an average of 80 percent for all letters with the Logistic Regression model with my own hand. So we decided to proceed further with the Logistic Regression model even though it did not have the highest accuracy on the validation set.

Despite the progress made, challenges remain, particularly in distinguishing between similar signs. Future work could focus on refining feature extraction techniques, adding support for left and right hands, expanding the dataset to include a broader variety of hand shapes and orientations, and integrating facial expressions and movements for a more comprehensive ASL interpretation.

Our research contributes a promising solution to enhance communication for individuals using ASL, highlighting the potential of machine learning and computer vision technologies to bridge language barriers and foster inclusivity in society.

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VIII. References

[1] S. Gupta, S. Bagga, and D. K. Sharma, "Hand Gesture Recognition for Human-Computer Interaction and Its Applications in Virtual Reality," in *Advanced Computational Intelligence Techniques for Virtual Reality in Healthcare*, **vol. 875**, D. Gupta, A. Hassanien, and A. Khanna, Eds. Cham: Springer, 2020, pp. 97-110. doi: 10.1007/978-3-030-35252-3_5.

[2] F. Karray, M. Alemzadeh, J. A. Saleh, and M. N. Arab, "Human-Computer Interaction: Overview on State of the Art," *Int. J. Smart Sens. Intell. Syst.*, **vol. 1, no. 1**, pp. 137-159 (2008). doi: 10.21307/ijssis-2017-283.

[3] J. Wang, W. Lin, and Z. Chen, "Smart touchless human–machine interaction based on crystalline porous cages," *Nat. Commun.*, **vol. 15, no. 1575**, pp. 1-9 (2024). doi: 10.1038/s41467-024-46071-8.

[4] Zhou, Z., Tam, V. W. L., Lam, E. Y. "A cross-attention BERT-based framework for continuous sign language recognition." *IEEE Signal Processing Letters*, **29**, 1818–1822.(2022) https://doi.org/10.1109/LSP.2022.3199665 [Google Scholar] [CrossRef]

[5] Luqman, H., El-Alfy, E. S. M. "Towards hybrid multimodal manual and non-manual Arabic sign language recognition: mArSL database and pilot study. *Electronics*, 10(14),1739(2021). https://doi.org/10.3390/electronics10141739 [Google Scholar] [CrossRef]

[6] S. Malassiotis, N. Aifanti and M. G. Strintzis, "A gesture recognition system using 3D data," Proceedings. *First International Symposium on 3D Data Processing Visualization and Transmission*, Padua, Italy(2002), pp. 190-193, doi: 10.1109/TDPVT.2002.1024061.

[7] Liddell, S.K. *American Sign Language Syntax*; Walter de Gruyter GmbH & Co KG: Berlin, Germany, 2021; **Volume 52**. [Google Scholar]

[8] Canese, L., Cardarilli, G. C., Di Nunzio, L., Fazzolari, R., Ghadakchi, H. F. et al. (2022). "Sensing and detection of traffic signs using CNNs: An assessment on their performance." *Sensors, 22(22)*, 8830. https://doi.org/10.3390/s22228830 [Google Scholar] [PubMed] [CrossRef]

[9] Manoharan, Y., Saxena, S., D., R. "A vision-based smart human computer interaction system for hand-gestures recognition." *1st International Conference on Computational Science and Technology (ICCST)*, pp. 321–324(2022). Sharjah, United Arab Emirates. https://doi.org/10.1109/ICCST55948.2022.10040464 [Google Scholar] [CrossRef]

[10] Hmida, I., Romdhane, N. B. "Arabic sign language recognition algorithm based on deep learning for smart cities." *The 3rd International Conference on Distributed Sensing and Intelligent Systems (ICDSIS)*, pp. 119–127(2022). Sharjah, United Arab Emirates. https://doi.org/10.1049/icp.2022.2426 [Google Scholar] [CrossRef]

[11] Starner, T.; Weaver, J.; Pentl, A. "Real-time american sign language recognition using desk and wearable computer based video." *IEEE Trans. Pattern Anal. Mach. Intell.*, *20*, 1371–1375(1998). [**Google Scholar**] [**CrossRef**]

[12] Ye, Y.; Tian, Y.; Huenerfauth, M.; Liu, J. "Recognizing american sign language gestures from within continuous

videos.” In Proceedings of the *IEEE Conference on Computer Vision and Pattern Recognition Workshops*, Salt Lake City, UT, USA, 18–23 (June 2018); pp. 2064–2073. [**Google Scholar**]

[13] K. -p. Feng and F. Yuan, "Static hand gesture recognition based on HOG characters and support vector machines," *2nd Int. Symp. Instrum. Meas., Sens. Netw. Autom. (IMSNA)*, Toronto, ON, Canada, (2013), pp. 936-938. doi: 10.1109/IMSNA.2013.6743432.

[14] V. Chang, R. O. Eniola, L. Golightly, et al., "An Exploration into Human–Computer Interaction: Hand Gesture Recognition Management in a Challenging Environment," *SN Comput. Sci.*, **vol. 4**, p. 441, (2023). doi: 10.1007/s42979-023-01751-y.

[15] Avina, V.D.; Amiruzzaman, M.; Amiruzzaman, S.; Ngo, L.B.; Dewan, M.A.A. “An AI-Based Framework for Translating American Sign Language to English and Vice Versa.”*Information*,*14*,569(2023). https://doi.org/10.3390/info14100569

[16] Grassknoted, "ASL Alphabet Dataset," *Kaggle*, 2024. [Online]. Available: https://www.kaggle.com/datasets/grassknoted/asl-alphabet

[17] N. Kansara, A. Rajkumar, P. Dharmale and M. Khadtare, "Enhancing Autism Spectrum Disorder Recognition in EEG Data through Filtering-Driven CNN Approach," *2024 IEEE International Conference on Consumer Electronics (ICCE)*, Las Vegas, NV, USA, (2024), pp. 1-7, doi: 10.1109/ICCE59016.2024.10444503.

[18] Khadtare, M.S., Sahambi, J.S. “ECG Arrhythmia Analysis by Multicategory Support Vector Machine”. *Applied Computing. AACC* (2004). Lecture Notes in Computer Science, vol 3285. Springer, Berlin, Heidelberg.

[19] Khadtare, M. Dharmale P., “Enhanced Human computer interaction using hand gesture analysis” on GPU Poster · (May 2015) DOI: 10.13140/RG.2.2.22476.62089